

RPG Character & Equipment System - UML Class Diagram

===== FACTORY METHOD =====

<<abstract>> CharacterFactory

+ orderCharacter(name)
createCharacter(name)

|
|-- WarriorFactory
|-- MageFactory
|-- ArcherFactory

<<interface>> Character

+ getName()
+ displayStats()
+ useSpecialAbility()
+ attack(enemy)
+ equipWeapon()
+ equipArmor()

|
|-- Warrior
|-- Mage
|-- Archer

===== ABSTRACT FACTORY =====

<<interface>> EquipmentFactory

+ createWeapon()
+ createArmor()

|
|-- MedievalFactory
|-- MagicFactory
|-- RangerFactory

<<interface>> Weapon

+ getDamage()
+ getWeaponInfo()
+ displayInfo()

|
|-- IronSword
|-- WizardStaff
|-- Longbow

<<interface>> Armor

+ getDefense()
+ getArmorInfo()
+ displayInfo()

|
|-- PlateArmor
|-- EnchantedRobes
|-- LeatherArmor

===== RELATIONSHIPS =====

Character "has-a" Weapon

Character "has-a" Armor

CharacterFactory --> Character

EquipmentFactory --> Weapon

EquipmentFactory --> Armor